

Parcial 2-RDC

Fidel Ignacio Peralta Rivas

Ejercicio 1: Red principal utilizada 10.31.75.0/24

Tabla de direccionamiento VLSM

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Red personalizada: A: 19, B: 63

10.31.75.0/24

Link 1: S0	Link 2: S5	Link 3: 12	Link 4: 6	Link R1-R2	Link R2-R3
2-2-200	2-2-205	2-2-12	2-2-24	2-2-22	
6-2-206	30-205	1-2-12	6-2-6	2-2-2	
2-2-206	2-2-205/20	2-2-20/28	2-2-20/24	2-2-20/20	
255.255.255.192	255.255.255.224	255.255.255.240	255.255.255.248	255.255.255.252	255.255.255.252
0.0.0.63	0.0.0.31	0.0.0.15	0.0.0.7	0.0.0.3	0.0.0.3
Salto: 64	Salto: 32	Salto: 16	Salto: 8	Salto: 4	Salto: 4

Tabla VLSM

area	dir sub	cidr	mask	1 host	ultima	direccion
lan 1	10.31.75.0	/26	255.255.255.192	10.31.75.1	10.31.75.62	10.31.75.63
lan 2	10.31.75.64	/27	255.255.255.224	10.31.75.65	10.31.75.94	10.31.75.95
lan 3	10.31.75.96	/28	255.255.255.240	10.31.75.97	10.31.75.110	10.31.75.111
lan 4	10.31.75.112	/29	255.255.255.248	10.31.75.113	10.31.75.118	10.31.75.119
R1-R2	10.31.75.120	/30	255.255.255.252	10.31.75.121	10.31.75.122	10.31.75.123
R2-R3	10.31.75.124	/30	255.255.255.252	10.31.75.125	10.31.75.126	10.31.75.127

Tabla de direccionamiento

area	direccion	interfaz	dir 1/1P	mask	gateway
lan 1	PC1	NIC	10.31.75.2	255.255.255.192	10.31.75.1
	R1	FE0/0	10.31.75.1	C+	N/A
lan 2	PC2	NIC	10.31.75.65	255.255.255.224	10.31.75.65
	R1	FE0/1	10.31.75.65	C+	N/A
lan 3	PC3	NIC	10.31.75.98	255.255.255.240	10.31.75.97
	R1	FE1/0	10.31.75.97	C+	N/A
lan 4	PC4	NIC	10.31.75.114	255.255.255.248	10.31.75.113
	R3	FE0/0	10.31.75.113	C+	N/A
R1-R2	R1	FE0/0	10.31.75.121	255.255.255.252	N/A
	R2	FE0/0	10.31.75.122		N/A
R2-R3	R2	FE0/0	10.31.75.125	255.255.255.252	N/A
	R3	FE0/1	10.31.75.126		N/A

Configuraciones de vPC's desde vpc1 hasta vp4

```
PC1 R1 PC2 PC3 PC4 R3 R2
84 bytes from 10.31.75.1 icmp_seq=3 ttl=255 time=15.833 ms
84 bytes from 10.31.75.1 icmp_seq=4 ttl=255 time=16.097 ms

PC1> ping 10.31.75.121
84 bytes from 10.31.75.121 icmp_seq=1 ttl=255 time=15.034 ms
84 bytes from 10.31.75.121 icmp_seq=2 ttl=255 time=15.964 ms
84 bytes from 10.31.75.121 icmp_seq=3 ttl=255 time=18.079 ms
84 bytes from 10.31.75.121 icmp_seq=4 ttl=255 time=16.071 ms
84 bytes from 10.31.75.121 icmp_seq=5 ttl=255 time=16.288 ms

PC1> ping 10.31.75.122
10.31.75.122 icmp_seq=1 timeout
10.31.75.122 icmp_seq=2 timeout
10.31.75.122 icmp_seq=3 timeout
10.31.75.122 icmp_seq=4 timeout
10.31.75.122 icmp_seq=5 timeout

PC1> ping 10.31.75.114
10.31.75.114 icmp_seq=1 timeout
10.31.75.114 icmp_seq=2 timeout
84 bytes from 10.31.75.114 icmp_seq=3 ttl=61 time=77.070 ms
84 bytes from 10.31.75.114 icmp_seq=4 ttl=61 time=78.634 ms
84 bytes from 10.31.75.114 icmp_seq=5 ttl=61 time=76.960 ms

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 10.31.75.2/26
GATEWAY    : 10.31.75.1
DNS        :
MAC        : 00:50:79:66:68:02
LPORT      : 10000
RHOST:PORT : 127.0.0.1:10001
MTU        : 1500

PC1> █
```

```
PC1 PC2 PC3 PC4 R3 R2
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...
PC1 : 10.31.75.66 255.255.255.224 gateway 10.31.75.65

PC2> ping 10.31.75.65
10.31.75.65 icmp_seq=1 timeout
84 bytes from 10.31.75.65 icmp_seq=2 ttl=255 time=16.314 ms
84 bytes from 10.31.75.65 icmp_seq=3 ttl=255 time=15.909 ms
84 bytes from 10.31.75.65 icmp_seq=4 ttl=255 time=15.454 ms
84 bytes from 10.31.75.65 icmp_seq=5 ttl=255 time=15.863 ms

PC2> ping 10.31.75.2
84 bytes from 10.31.75.2 icmp_seq=1 ttl=63 time=30.738 ms
84 bytes from 10.31.75.2 icmp_seq=2 ttl=63 time=31.064 ms
84 bytes from 10.31.75.2 icmp_seq=3 ttl=63 time=30.905 ms
84 bytes from 10.31.75.2 icmp_seq=4 ttl=63 time=31.751 ms
84 bytes from 10.31.75.2 icmp_seq=5 ttl=63 time=31.900 ms

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 10.31.75.66/27
GATEWAY    : 10.31.75.65
DNS        :
MAC        : 00:50:79:66:68:03
LPORT      : 10002
RHOST:PORT : 127.0.0.1:10003
MTU        : 1500

PC2> █
```

```

Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
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VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...
PC1 : 10.31.75.98 255.255.255.240 gateway 10.31.75.97

PC3> show ip

NAME       : PC3[1]
IP/MASK     : 10.31.75.98/28
GATEWAY     : 10.31.75.97
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 10004
RHOST:PORT  : 127.0.0.1:10005
MTU         : 1500

PC3>

```

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```

PC4> trace 10.31.75.2
trace to 10.31.75.2, 8 hops max, press Ctrl+C to stop
 1  10.31.75.113  15.085 ms  14.458 ms  14.620 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *

 8  * * *

PC4>
PC4> ping 10.31.75.125
10.31.75.125 icmp_seq=1 timeout
10.31.75.125 icmp_seq=2 timeout
10.31.75.125 icmp_seq=3 timeout
10.31.75.125 icmp_seq=4 timeout
10.31.75.125 icmp_seq=5 timeout

PC4> show ip

NAME       : PC4[1]
IP/MASK     : 10.31.75.114/29
GATEWAY     : 10.31.75.113
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 10025
RHOST:PORT  : 127.0.0.1:10026
MTU         : 1500

PC4>

```

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Configuración de las interfaces y configuración de rutas estaticas en cada router, empezando por R1 en orden hasta R3

```
R1#show ip routes
^
% Invalid input detected at '^' marker.

R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
C       10.31.75.64/27 is directly connected, FastEthernet0/1
S       10.31.75.124/30 [1/0] via 10.31.75.122
C       10.31.75.120/30 is directly connected, Serial0/0
S       10.31.75.112/29 [1/0] via 10.31.75.122
C       10.31.75.96/28 is directly connected, FastEthernet1/0
C       10.31.75.0/26 is directly connected, FastEthernet0/0
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.31.75.1      YES NVRAM   up          up
Serial0/0                 10.31.75.121    YES manual  up          up
FastEthernet0/1          10.31.75.65     YES NVRAM   up          up
Serial0/1                 unassigned      YES unset   administratively down down
Serial0/2                 unassigned      YES unset   administratively down down
Serial0/3                 unassigned      YES unset   administratively down down
FastEthernet1/0          10.31.75.97     YES NVRAM   up          up
FastEthernet2/0          unassigned      YES unset   up          up
FastEthernet2/1          unassigned      YES unset   up          down
FastEthernet2/2          unassigned      YES unset   up          down
FastEthernet2/3          unassigned      YES unset   up          down
--More--
```

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```

*Mar  1 00:54:40.187: %SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
    S    10.31.75.64/27 [1/0] via 10.31.75.121
    C    10.31.75.124/30 is directly connected, FastEthernet0/0
    C    10.31.75.120/30 is directly connected, Serial1/0
    S    10.31.75.112/29 [1/0] via 10.31.75.126
    S    10.31.75.96/27 [1/0] via 10.31.75.121
    S    10.31.75.0/26 [1/0] via 10.31.75.121
R2#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
FastEthernet0/0          10.31.75.125    YES manual up                    up
FastEthernet0/1          unassigned      YES unset  administratively down  down
Serial1/0                 10.31.75.122    YES manual up                    up
Serial1/1                 unassigned      YES unset  administratively down  down
Serial1/2                 unassigned      YES unset  administratively down  down
Serial1/3                 unassigned      YES unset  administratively down  down
FastEthernet2/0           unassigned      YES unset up                    up
FastEthernet2/1           unassigned      YES unset up                    down
FastEthernet2/2           unassigned      YES unset up                    down
FastEthernet2/3           unassigned      YES unset up                    down
FastEthernet2/4           unassigned      YES unset up                    down
--More--
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```

```

S    10.31.75.120/30 [1/0] via 10.31.75.125
C    10.31.75.112/29 is directly connected, FastEthernet0/0
S    10.31.75.96/28 [1/0] via 10.31.75.125
S    10.31.75.0/26 [1/0] via 10.31.75.125
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
    S    10.31.75.64/27 [1/0] via 10.31.75.125
    C    10.31.75.124/30 is directly connected, FastEthernet0/1
    S    10.31.75.120/30 [1/0] via 10.31.75.125
    C    10.31.75.112/29 is directly connected, FastEthernet0/0
    S    10.31.75.96/28 [1/0] via 10.31.75.125
    S    10.31.75.0/26 [1/0] via 10.31.75.125
R3#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
FastEthernet0/0          10.31.75.113    YES manual up                    up
FastEthernet0/1          10.31.75.126    YES manual up                    up
FastEthernet1/0           unassigned      YES unset  administratively down  down
FastEthernet2/0           unassigned      YES unset up                    down
FastEthernet2/1           unassigned      YES unset up                    down
FastEthernet2/2           unassigned      YES unset up                    down
FastEthernet2/3           unassigned      YES unset up                    down
FastEthernet2/4           unassigned      YES unset up                    down
FastEthernet2/5           unassigned      YES unset up                    down
FastEthernet2/6           unassigned      YES unset up                    down
FastEthernet2/7           unassigned      YES unset up                    down
--More--
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```

Trace desde vPC1 hasta vPC4

```
PC1  R1  PC2  PC3  PC4  R3  R2
84 bytes from 10.31.75.114 icmp_seq=4 ttl=61 time=78.634 ms
84 bytes from 10.31.75.114 icmp_seq=5 ttl=61 time=76.960 ms

PC1> show ip

NAME       : PC1[1]
IP/MASK     : 10.31.75.2/26
GATEWAY     : 10.31.75.1
DNS         :
MAC         : 00:50:79:66:68:02
LPORT      : 10000
RHOST:PORT  : 127.0.0.1:10001
MTU         : 1500

PC1> ping 10.31.75.114
10.31.75.114 icmp_seq=1 timeout
10.31.75.114 icmp_seq=2 timeout
84 bytes from 10.31.75.114 icmp_seq=3 ttl=61 time=62.702 ms
84 bytes from 10.31.75.114 icmp_seq=4 ttl=61 time=78.698 ms
84 bytes from 10.31.75.114 icmp_seq=5 ttl=61 time=78.812 ms

PC1> ping 10.31.75.114
84 bytes from 10.31.75.114 icmp_seq=1 ttl=61 time=78.467 ms
84 bytes from 10.31.75.114 icmp_seq=2 ttl=61 time=63.258 ms
84 bytes from 10.31.75.114 icmp_seq=3 ttl=61 time=80.895 ms
84 bytes from 10.31.75.114 icmp_seq=4 ttl=61 time=77.591 ms
84 bytes from 10.31.75.114 icmp_seq=5 ttl=61 time=63.049 ms

PC1> trace 10.31.75.114
trace to 10.31.75.114, 8 hops max, press Ctrl+C to stop
 1  10.31.75.1    15.386 ms  15.776 ms  15.624 ms
 2  10.31.75.122  30.919 ms  29.491 ms  29.849 ms
 3  10.31.75.126  60.482 ms  61.503 ms  45.829 ms
 4  *10.31.75.114 76.016 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> █
```

Trace desde vPC4 hasta vPC2

```

10.31.75.125 icmp_seq=4 timeout
10.31.75.125 icmp_seq=5 timeout

PC4> show ip

NAME       : PC4[1]
IP/MASK     : 10.31.75.114/29
GATEWAY     : 10.31.75.113
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 10025
RHOST:PORT  : 127.0.0.1:10026
MTU         : 1500

PC4> ping 10.31.75.66
10.31.75.66 icmp_seq=1 timeout
10.31.75.66 icmp_seq=2 timeout
84 bytes from 10.31.75.66 icmp_seq=3 ttl=61 time=76.785 ms
84 bytes from 10.31.75.66 icmp_seq=4 ttl=61 time=78.831 ms
84 bytes from 10.31.75.66 icmp_seq=5 ttl=61 time=77.376 ms

PC4> ping 10.31.75.66
84 bytes from 10.31.75.66 icmp_seq=1 ttl=61 time=77.503 ms
84 bytes from 10.31.75.66 icmp_seq=2 ttl=61 time=50.109 ms
84 bytes from 10.31.75.66 icmp_seq=3 ttl=61 time=76.803 ms
84 bytes from 10.31.75.66 icmp_seq=4 ttl=61 time=64.937 ms
84 bytes from 10.31.75.66 icmp_seq=5 ttl=61 time=79.503 ms

PC4> trace 10.31.75.66
trace to 10.31.75.66, 8 hops max, press Ctrl+C to stop
 1  10.31.75.113  16.587 ms  14.889 ms  15.482 ms
 2  10.31.75.125  45.172 ms  45.708 ms  46.274 ms
 3  10.31.75.121  63.458 ms  47.667 ms  59.205 ms
 4  *10.31.75.66  77.055 ms (ICMP type:3, code:3, Destination port unreachable)

PC4> █
```

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